

ML Foundations — Day 2: Real Supervised Models & Preprocessing

Dataset: Adult / Census Income (UCI, via OpenML) | Qasim | random_state = 42

Preprocessing Choices & Rationale

Numeric features (age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week) are median-imputed then standardized; categorical features (workclass, education, marital-status, occupation, relationship, race, sex, native-country) are most-frequent-imputed then one-hot encoded with **handle_unknown='ignore'**. Median imputation was chosen over mean because capital-gain and capital-loss are heavily right-skewed with a small number of extreme values, which would pull the mean away from a representative "typical" value. One-hot encoding was chosen over label/ordinal encoding because none of these categories have a natural order (e.g. no ranking between workclass values), and label encoding would falsely imply one. Target encoding was considered but rejected at this stage due to leakage risk if not carefully cross-validated. Both preprocessing branches are combined in a single **ColumnTransformer**, wrapped with each estimator in one **Pipeline**, and fitted only on the 70% training split — the held-out test set was never used to fit any imputer, scaler, or encoder.

Model Performance Comparison (held-out test set, n = 9,769)

Model	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Day 1 Baseline B (edu-num $\geq$ 13)	0.7530	0.4844	0.4970	0.4906	0.7189	0.4443
Logistic Regression	0.8527	0.7431	0.5877	0.6563	0.9039	0.7624
Decision Tree (unconstrained)	0.8183	0.6192	0.6253	0.6223	0.7522	0.4769

Both supervised models clearly beat the Day 1 rule-based baseline. Logistic Regression is the strongest model overall (ROC AUC 0.90 vs. 0.75 for the tree and 0.72 for the baseline; PR AUC 0.76 vs. 0.48/0.44). The unconstrained Decision Tree reached higher recall than Logistic Regression (0.63 vs. 0.59) but at a large precision cost (0.62 vs. 0.74) and, more importantly, shows severe overfitting: **99.99% training accuracy vs. 81.83% test accuracy** (an 18-point gap) at a fitted depth of 56. For both models, **false negatives outnumber false positives** (Logistic Regression: 475 FP / 964 FN; Decision Tree: 899 FP / 876 FN), a direct consequence of class imbalance — the minority (>50K) class is more often under-predicted. Operationally, false positives cost the campaign wasted outreach while false negatives are a missed-opportunity cost; because the primary metric (ROC AUC) is threshold-independent, the campaign team can shift the decision threshold post-hoc to trade recall for precision depending on which error type they most want to avoid, without needing to retrain the model.

Interpretability Insights

Logistic Regression — top positive coefficients: capital-gain (~2.22, by far the strongest single feature), marital-status_Married-civ-spouse (~1.59), marital-status_Married-AF-spouse (~1.22), occupation_Exec-managerial (~0.79), relationship_Wife (~0.76), and education-num (~0.76) — all align with well-known income correlates (capital income, marriage/dual-income households, managerial roles, schooling).

Top negative coefficients: occupation_Priv-house-serv (~-1.41), marital-status_Never-married (~-1.23), sex_Female (~-1.00), relationship_Other-relative (~-0.95), occupation_Farming-fishing (~-0.92), and occupation_Other-service (~-0.86). The occupation/marital-status/education directions are intuitive; the sex_Female coefficient reflects a real, well-documented gender income gap in this 1994 census data and is flagged here as a fairness consideration to revisit before any real targeting use.

Decision Tree: top splits key off marital-status_Married-civ-spouse, capital-gain, and education-num — the same variables dominating the logistic regression, a useful cross-model consistency check. But at unconstrained depth 56 with a 18-point train/test accuracy gap, only the top 2–3 levels are trustworthy; deeper splits are likely memorizing individual training rows rather than generalizing.

Model Selection for Day 3

Logistic Regression is the model to carry into Day 3. It leads on both ROC AUC (0.90) and PR AUC (0.76), shows no overfitting gap, and its coefficients map directly onto interpretable feature effects (Task 4) — valuable for a business stakeholder who needs to understand **why** a household is flagged. The Decision Tree remains useful diagnostically: its top splits corroborate the logistic regression's most important features, but its unconstrained depth makes it unreliable as-is and it would need explicit regularization (`max_depth`, `min_samples_leaf`, or cost-complexity pruning) before it could be considered a serious candidate.

The fitted preprocessing `ColumnTransformer` is saved (pickled) for reuse without modification. Planned changes for Day 3: (1) regularize the Decision Tree and re-compare; (2) group rare native-country/occupation levels into an "Other" bucket before one-hot encoding; (3) try L1-regularized logistic regression for automatic feature selection; (4) test class-weighting or threshold tuning, since false negatives currently dominate and the business may prefer to shift that balance.